

Amos Roche

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EDUCATION

Columbia University Sep 2023 - Dec 2024
Master of Science in Data Science New York City, NY

Worcester Polytechnic Institute Aug 2020 - May 2023
Bachelor of Science in Data Science Worcester, MA

WORK EXPERIENCE

Kaliber Labs Mar 2025 - Present
ML Engineer San Francisco, CA

- Led end-to-end development of a **TTS System**, assembling **10M+ speech clips** and implementing **conditional flow-matching** and **state-space** architectures to improve expressiveness and naturalness.
- Developed a **hierarchical multi-task learning** architecture using **WavLM** and **cross-attention**, achieving state-of-the-art (SOTA) performance on L2-ARCTIC with **13.47% phoneme error rate** through articulatory feature supervision; manuscript in preparation.
- Built and productionized an **AI document-intake pipeline** for hospital systems serving **300+ patients** and **50+ providers**.
- Designed **context-engineering pipelines for a 4-bit quantized LLM on edge hardware**, structuring conversation history and patient-state data to drive digital-twin reasoning and updates.

Walmart Connect Jun 2024 - Aug 2024
Data Science Intern (Ads) New York City, NY

- Identified **control/test-group mismatches in ~13% of ad campaigns** using A/A testing and correlation analysis, driving updates to campaign shareability criteria and preventing inaccurate sales-lift reporting to advertisers.
- Built **PySpark/SQL data pipelines** to automate daily time-series ingestion from Hive to GCP, improving data-pull performance by 7.8%

Fidelity Investments Jun 2023 - Aug 2023
Data Engineering Intern Durham, NC

- Built Python/Groovy pipelines from **AWS S3** to **Snowflake** and migrated transformations from **ETL to dbt-based ELT**, improving efficiency by 6.7%; automated access and 30-day retention policies with **CloudFormation**

Fidelity Investments (Fidelity Center for Applied Technology) Jun 2022 - Aug 2022
Data Science Intern Boston, MA

- Developed an **NLP email-classification pipeline for 60+ business units**, improving performance for **55/60 units** using Linear SVC and advanced sampling techniques; built a **Flask front end** for comparing embeddings, sampling strategies, and model configurations

National Science Foundation Data Science REU May 2021 - Jan 2022
Machine Learning Researcher Worcester, MA

- Developed a **dynamic model-stacking framework** using convex optimization that outperformed baselines on **6/9 behavioral outcomes** for childhood cancer survivors; built a neural-network-based feature selection.

RESEARCH

Undergraduate Capstone Project - Aerial Image Processing Aug 2022 - May 2023
Computer Vision Researcher Worcester, MA

- Developed an **end-to-end computer vision pipeline for GPS-denied aerial localization**, training a **CNN-based regression model** to estimate positional shifts from parachute imagery.
- Applied **SegFormer transformer-based semantic segmentation** and **OpenCV contour/edge detection** to perform real-time terrain classification and extract geographic boundaries from aerial video.

PUBLICATIONS

Riaz, S., Zheng, H., **Roche, A.**, Zhang, M., Ragu, A., Penachio, S., Mukherjee, K., & Jonelagadda, A. (2026). *Multi-Task Learning for Non-Canonical Phoneme Recognition via Articulatory Feature Decomposition*. arXiv preprint arXiv:2608.22273.

Markus, A., **Roche, A.**, Ngan, C.-K., Cheung, Y.-T., & Prifti, K. (2022). *A Prognostic Machine Learning Framework and Algorithm for Predicting Long-term Behavioural Outcomes in Cancer Survivors*. DOI: 10.5220/0010893700003123

SKILLS

Languages: Python, SQL, Java, R, Scala, C++

ML Frameworks / Libraries: PyTorch, TensorFlow, Scikit-learn, Hugging Face, OpenCV

ML Techniques: Transformers, Multi-Task Learning, RAG, Diffusion Models, State-Space Models, Cross-Attention, CNNs

Speech / Audio ML: TTS, WavLM, Phoneme Recognition, CTC, Conditional Flow Matching

ML Systems: Edge Inference, Quantization, GPU Training & Inference, CUDA, Model Optimization

Data / Cloud: PySpark, dbt, Snowflake, AWS, GCP, BigQuery